

DATA SCIENCE PORTFOLIO PROJECT

Customer Shopping Behavior & Analytics Report

An exhaustive exploratory data analysis and database workflow uncovering commercial growth levers across 3,900 discrete consumer transactions.

CONFIDENTIAL EXECUTIVE DECK

ANALYTICS DIVISION

Dataset Structure & Engineering Pipeline

Rigorous data processing framework implemented prior to analytical evaluation.

DATASET SNAPSHOT

Core Metrics & Hygiene

PARAMETER	SPECIFICATION
Total Observations	3,900 Purchase Records
Attribute Matrix	18 Behavioral & Demographic Columns
Missing Data Handling	Median Imputation applied to 37 null Review Ratings
Storage Backend	PostgreSQL relational database schema

TRANSFORMATION FLOW

Pipeline Execution Steps

- **Ingestion:** Structured loading via Python/pandas runtime environment.
- **Feature Structuring:** Creation of uniform age buckets and granular purchase frequency matrices.
- **SQL Warehousing:** Indexed tables for efficient multi-variable queries and aggregation.
- **Validation:** Outlier checks performed across transaction amounts and regional distribution.

Revenue Drivers & Promotional Response

Cross-referencing demographic spend with incentive sensitivity.

DEMOGRAPHIC INSIGHT

Gender Revenue Distribution

Empirical evaluation indicates that female consumer segments yield marginally higher aggregate lifetime spend compared to male cohorts.

Commercial Takeaway:

Transition away from broad-stroke demographic advertising toward micro-targeted product recommendations tailored by behavioral clustering.

INCENTIVE ANALYSIS

The Discount Paradox

High-value spenders heavily leverage active promotional discounts to maximize cart value, rather than acting as low-margin bargain hunters.

Commercial Takeaway:

Restructure discount structures into tier-based VIP loyalty rewards instead of universal markdown models.

Logistics Impact & Subscription Value

Quantifying the correlation between premium service adoption and customer yield.

LOGISTICS BEHAVIOR

Shipping Tier Analysis

\$65

Express Shipping Basket

Average transaction size for priority delivery users.

\$58

Standard Shipping Basket

Average transaction size for standard delivery users.

Express shipping users exhibit a clear 12% higher order valuation per checkout.

SUBSCRIPTION ECONOMY

The Subscriber Advantage

- **+68% Higher Spend:** Active subscribers substantially outpace non-subscribers in baseline order volume.
- **45% Revenue Concentration:** Nearly half of total business revenue is anchored directly by subscription members.
- **78% Retention Rate:** Demonstrates high systemic repeat-purchase frequency across seasons.

Inventory Quality & Category Satisfaction

Isolating high-approval product lines that drive brand equity.

TIER 1 LEADER Blouse Category ★★★★★ Achieved the absolute highest customer review ratings across the entire catalog with minimal return rates.	TIER 1 LEADER Dress Category ★★★★★ Maintains stellar, flawless evaluations coupled with powerful multi-season repeat buyer interest.	TIER 2 VOLUME Shirt Category ★★★★☆ Sustains massive sales velocity backed by strong, consistent general market approval ratings.
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Customer Segmentation & Action Directives

Synthesizing behavioral metrics into an operational growth plan.

50% OF BASE

New Acquisition Cohort

First-time purchasers requiring structured, automated onboarding pathways to curb early churn.

35% OF BASE

Returning Shoppers

Consistent multi-category participants prime for cross-selling recommendations.

15% OF BASE

Loyal Advocates

High-value anchors driving reliable baseline revenue and organic brand promotion.